Swiss Institute of Bioinformatics

PR STATS

Metagenomic Data Analysis of Microbial Communities (MGMC01)

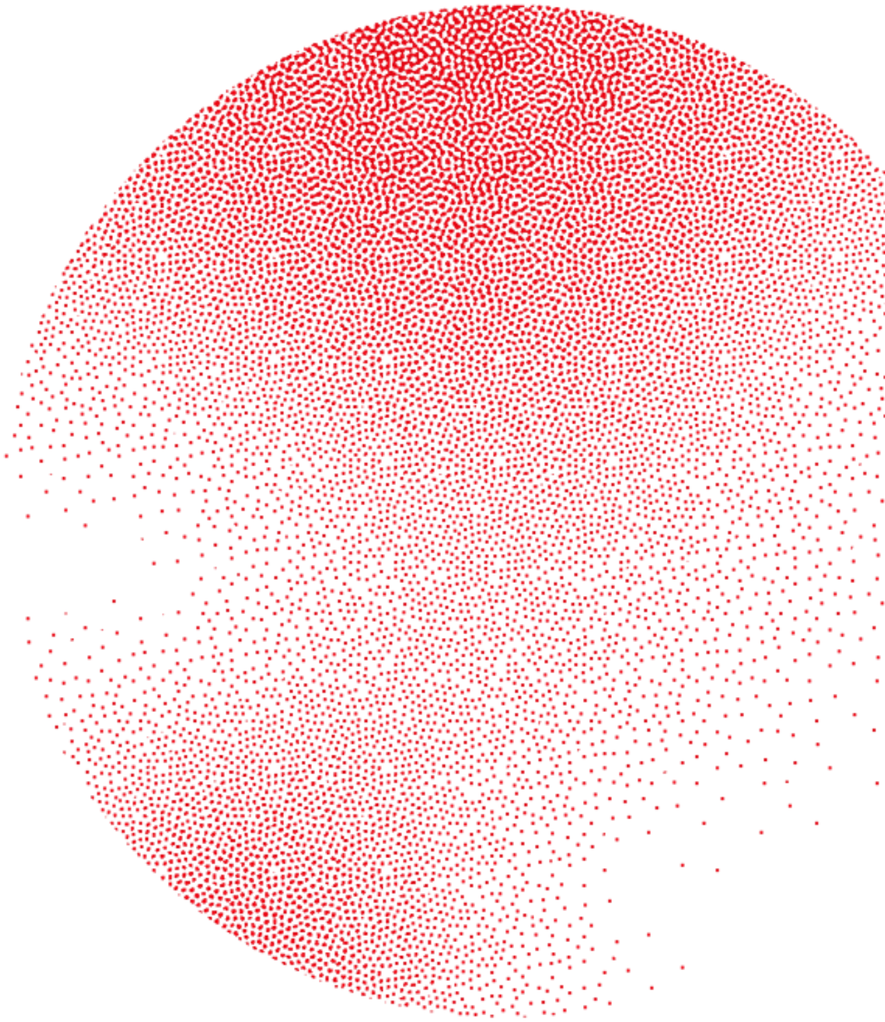
Day 3 - Session 2

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University of Fribourg
October 28th, 2026

elixir SWITZERLAND

This presentation

- Learning outcomes
- Downstream analysis
- BIOM and Phyloseq
- Diversity
- Compositionality
- Transformation and normalization
- Statistical modeling
- Functional profiling
- Summary



Learning outcomes

- Understand the downstream analysis as a critical step during taxonomic classification/profiling.
- Identify the different options available to provide an analysis as comprehensive as possible.
- Recognize the nature of metagenomics data in terms of its data structure.
- Perform the downstream analysis for the given dataset covering different aspects as diversity and statistical modeling.

Downstream analyses of classified reads

From Kraken2 and MetaPhlAn 4 outputs to biological interpretation

Kraken2 + Bracken · read counts **MetaPhlAn 4** · relative abundances

Complementary

1 Tables and visualisation

- Merged abundance table
- Filtering and normalisation
- Stacked bar plots
- Heatmaps and Krona charts

2 Community analysis

- α-diversity
- β-diversity
- Ordination (PCoA, NMDS)
- PERMANOVA

3 Differential analysis

- Differential abundance (MaAsLin2, ANCOM-BC)
- LEfSe
- Machine learning

4 Beyond taxonomy

- Functional profiling (HUMAnN)
- Strain-level analysis (StrainPhlAn)

BIOM files

Biological Observation Matrix: A standardized file format for representing biological count data (storage).

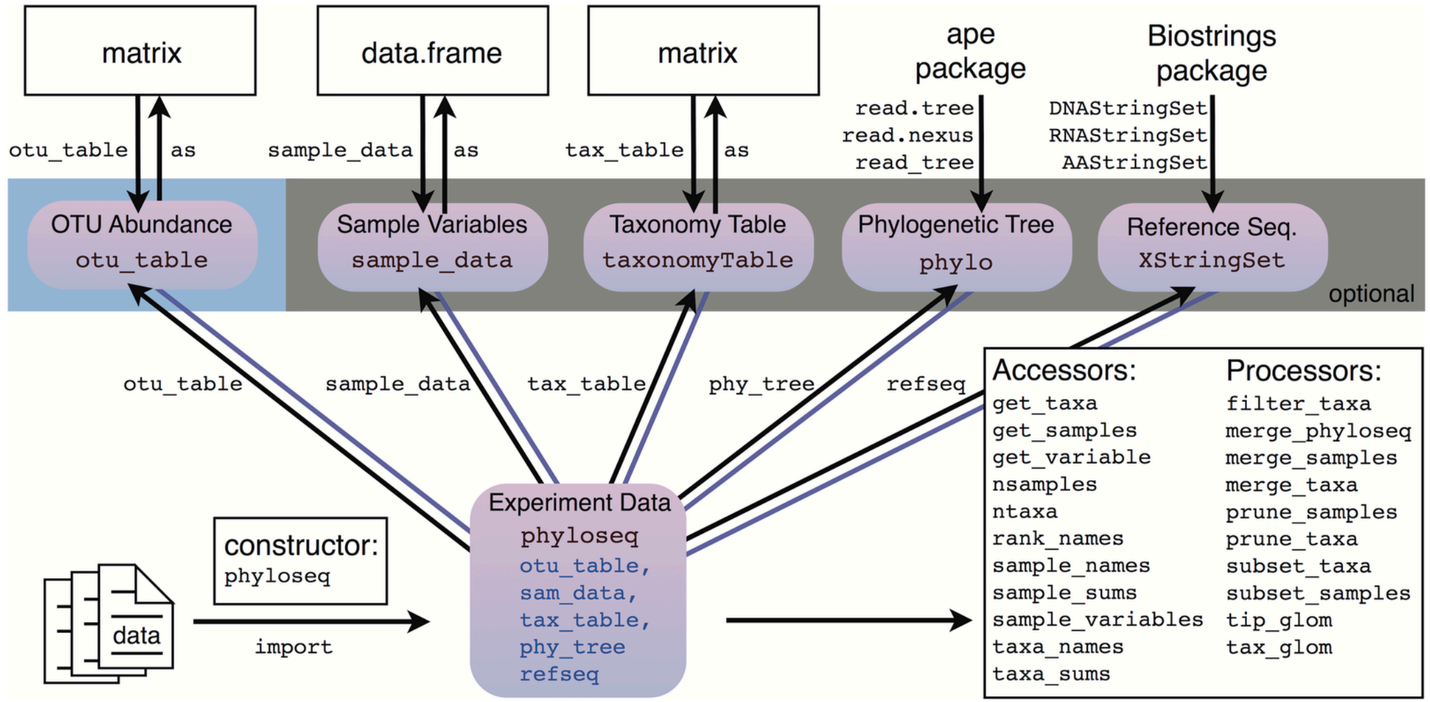
- Feature table.
- Metadata.
- Taxonomy.
- Optional: Tree

```
>>> import numpy as np
>>> from biom.table import Table
>>> data = np.arange(40).reshape(10, 4)
>>> sample_ids = ['S%d' % i for i in range(4)]
>>> observ_ids = ['O%d' % i for i in range(10)]
>>> sample_metadata = [{'environment': 'A'}, {'environment': 'B'},
...                    {'environment': 'A'}, {'environment': 'B'}]
>>> observ_metadata = [{'taxonomy': ['Bacteria', 'Firmicutes']},
...                    {'taxonomy': ['Bacteria', 'Firmicutes']},
...                    {'taxonomy': ['Bacteria', 'Proteobacteria']},
...                    {'taxonomy': ['Bacteria', 'Proteobacteria']},
...                    {'taxonomy': ['Bacteria', 'Proteobacteria']},
...                    {'taxonomy': ['Bacteria', 'Bacteroidetes']},
...                    {'taxonomy': ['Bacteria', 'Bacteroidetes']},
...                    {'taxonomy': ['Bacteria', 'Firmicutes']},
...                    {'taxonomy': ['Bacteria', 'Firmicutes']},
...                    {'taxonomy': ['Bacteria', 'Firmicutes']}]
>>> table = Table(data, observ_ids, sample_ids, observ_metadata,
...               sample_metadata, table_id='Example Table')
```

Phyloseq

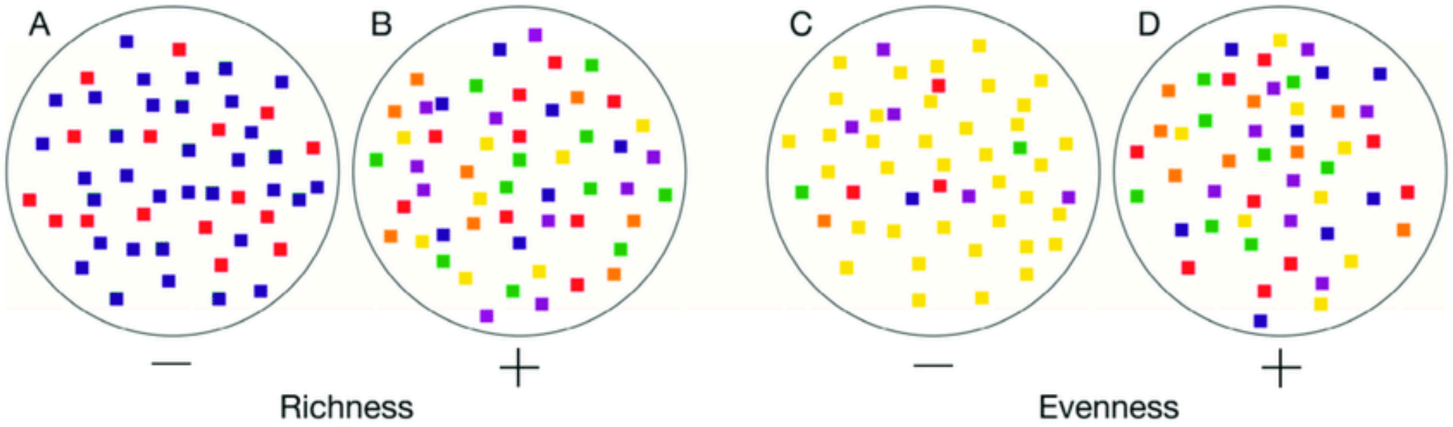
Phyloseq object: A structured container in R that integrates multiple data types (analysis):

- “OTU” abundance table.
- Taxonomy table.
- Sample metadata.
- Phylogenetic tree (optional).



Diversity analysis

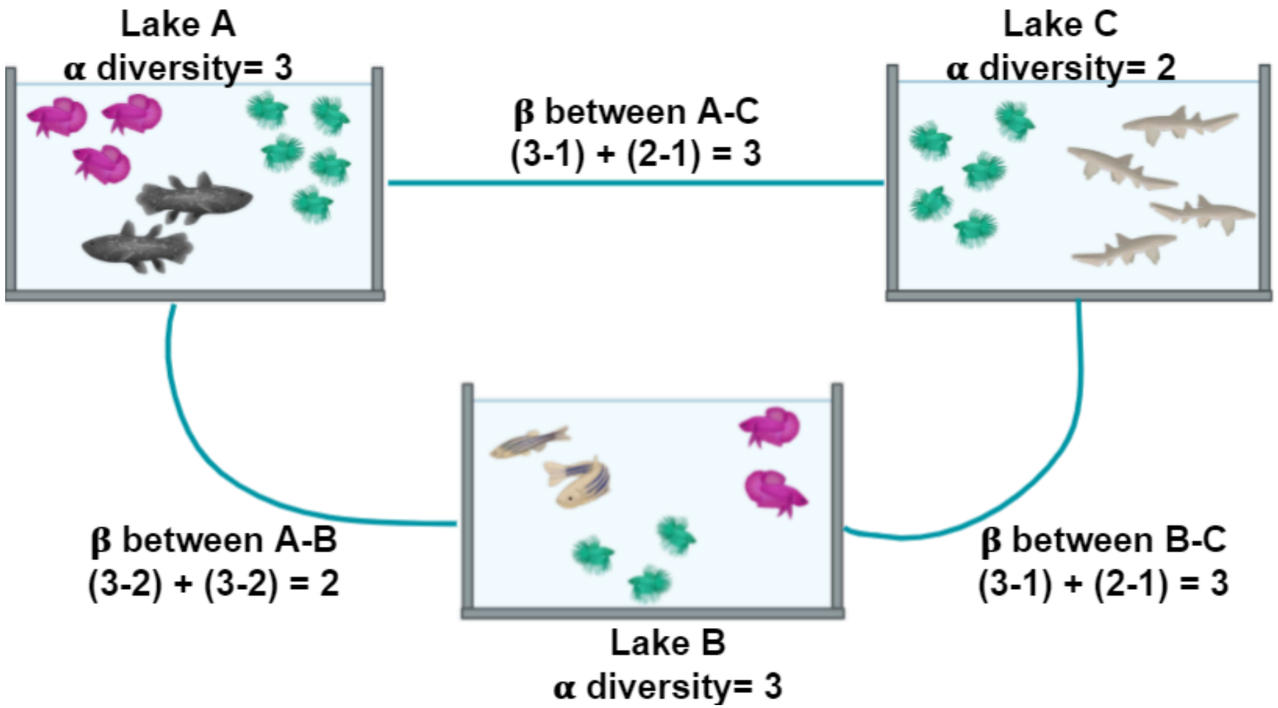
α-Diversity: Assess how complex or diverse an individual microbial community is.



Diversity Indices	Description
Shannon (H)	Estimation of species richness and species evenness. More weight on richness.
Simpson's (D)	Estimation of species richness and species evenness. More weigth on evenness.
Chao1	Abundance based on species represented by a single individual (singletons) and two individuals (doubletons).

Diversity analysis

β-Diversity: Measures differences in community composition between samples



- Bray-Curtis.

$$BC_d = \frac{\sum |x_i - x_j|}{\sum (x_i + x_j)}$$

- Jaccard.
- UniFrac.
- Weighted Unifrac.
- Manhattan.
- Euclidean.

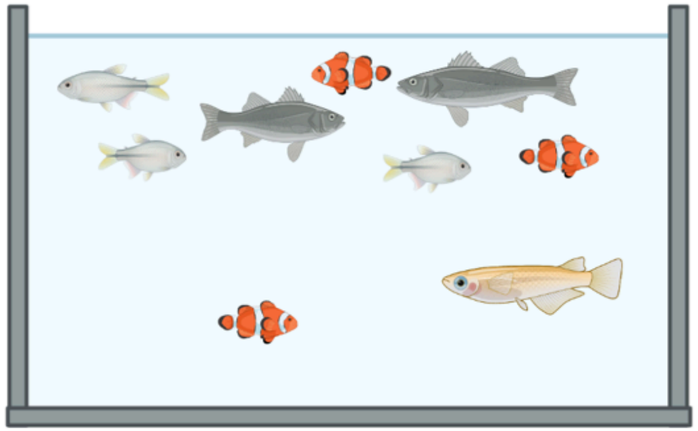
Zirión-Martínez et al. (2024)

Diversity analysis

Exercise:

Lake A

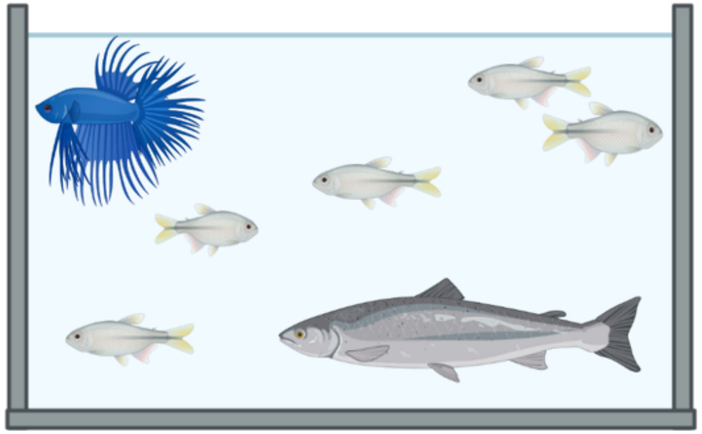
α-diversity= ?



A-B
β-diversity= ?

Lake B

α-diversity= ?



α-Diversity lake A?
α-Diversity lake B?
β-Diversity lake A-B?

Metagenomics data is compositional

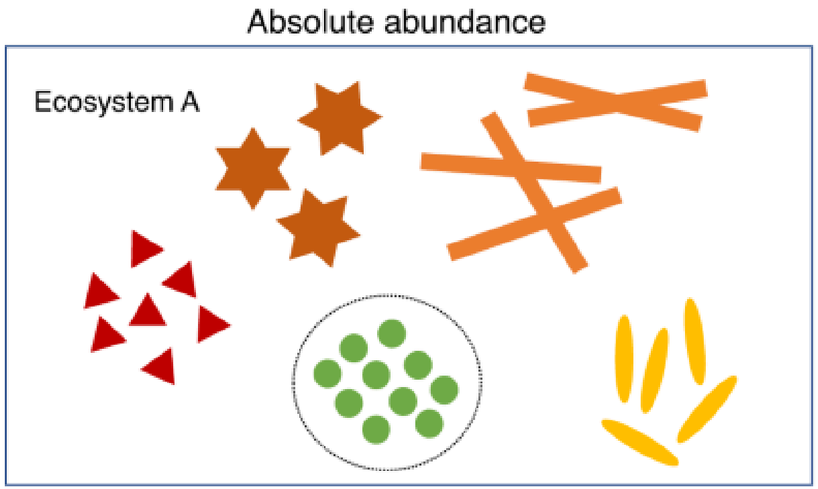
- The total number of counts per sample is highly variable and **constrained by the maximum number** of DNA reads that the sequencer can provide.

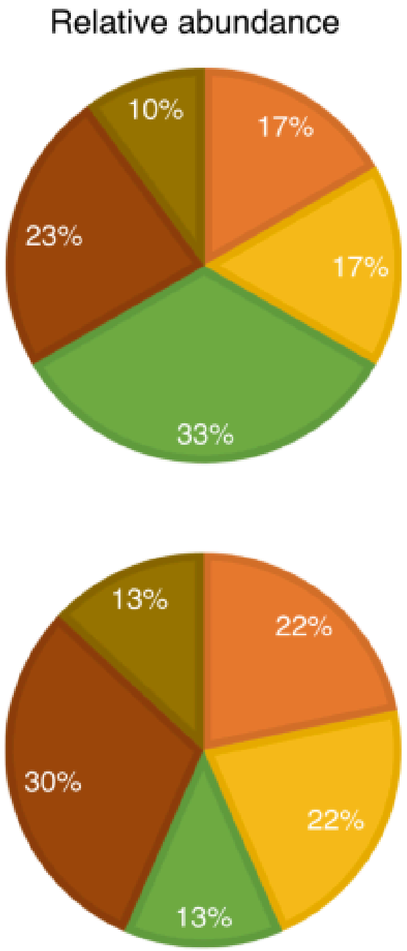
- An **increase** in the abundance of one taxon **implies** the **decrease** of the observed number of counts for some of the other taxa.

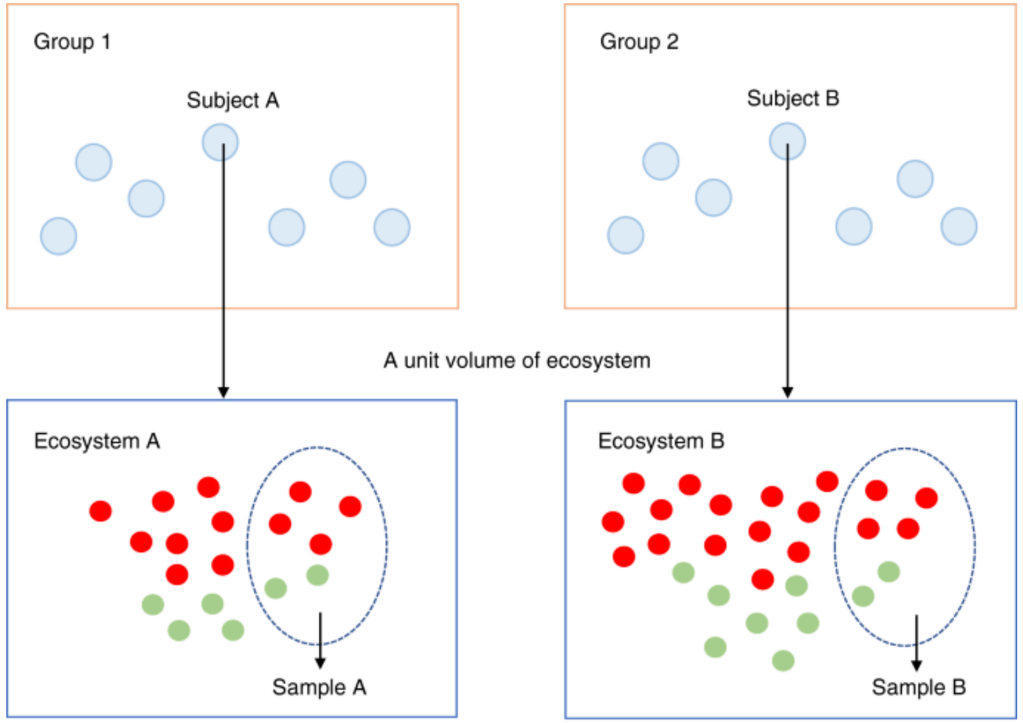
- Observed raw abundances and the total number of reads per sample are non-informative since they represent **only a fraction** or random sample of the **original DNA content in the environment**.

Calle (2019)

Metagenomics data is compositional







	Sample		Ecosystem	
	A	B	A	B
●	4	4	12	18
●	2	2	6	9
Sum	6†	6†	18‡	27‡

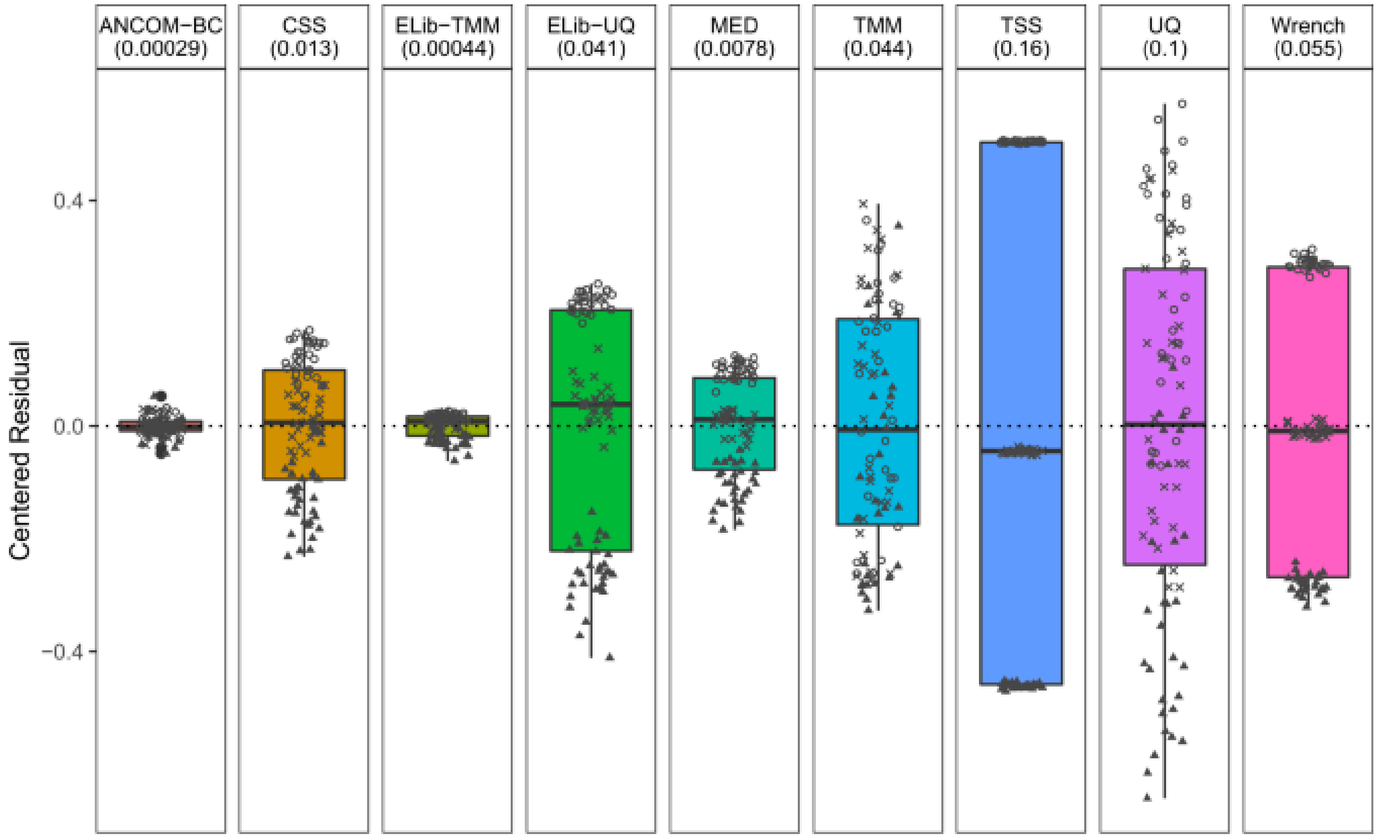
† Library size; ‡ microbial load.

[Lin & Peddada (2020)]

Bias introduced by sampling fraction

Data normalization

Normalization: Adjusts for differences in sequencing depth or library size



Lin & Peddada (2020)

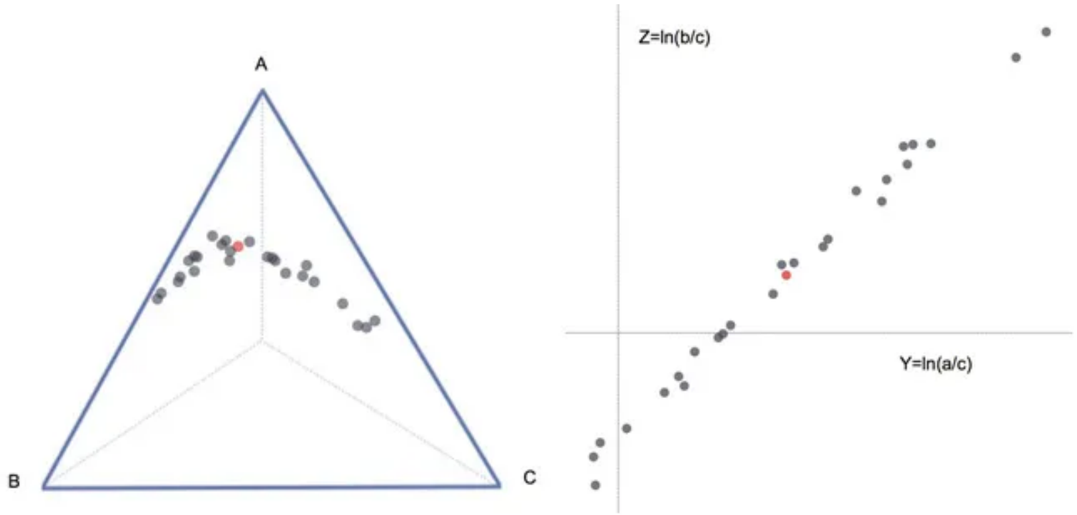
Operation	Standard approach	Compositional approach
Normalization	Rarefaction 'DESeq'	CLR ILR ALR
Distance	Bray-Curtis UniFrac Jenson-Shannon	Aitchison
Ordination	PCoA (Abundance)	PCA (Variance)
Multivariate comparison	perManova ANOSIM	perMANOVA ANOSIM
Correlation	Pearson Spearman	SparCC SpiecEasi ϕ ρ
Differential abundance	metagenomSeq LEfSe DESeq	ALDEx2 ANCOM

Gloor et al. (2017)

Transformations

Centered-log transformation*: Modifies data distribution (compositional)

$$clr(x_1, \cdots, x_k) = \left(\log\left(\frac{x_1}{g(x)}\right), \ldots, \log\left(\frac{xk}{g(x)}\right)\right)$$



Conditions to fulfill:

- Permutation invariance (**order**)
- Scale invariance (**multiplication by a factor**)
- Sub-compositional coherence (**presence/absence**)

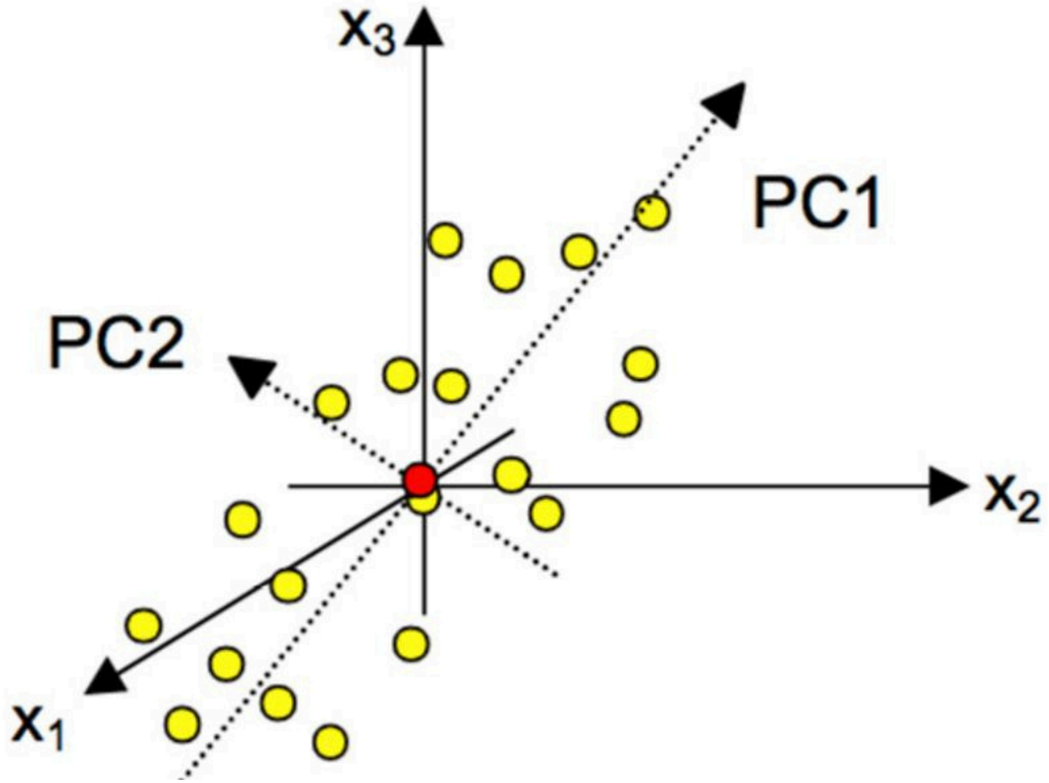
***Requires zero replacement.**

Ordination/Dimensionality reduction

Dimensionality reduction: Reduces high-dimensional data (many taxa/features) into fewer variables.

PCA: Uses as input the data matrix and models covariance (features driving the variation).

PCoA: Uses as input a distance/dissimilarity matrix and models pairwise distances (community level shifts).

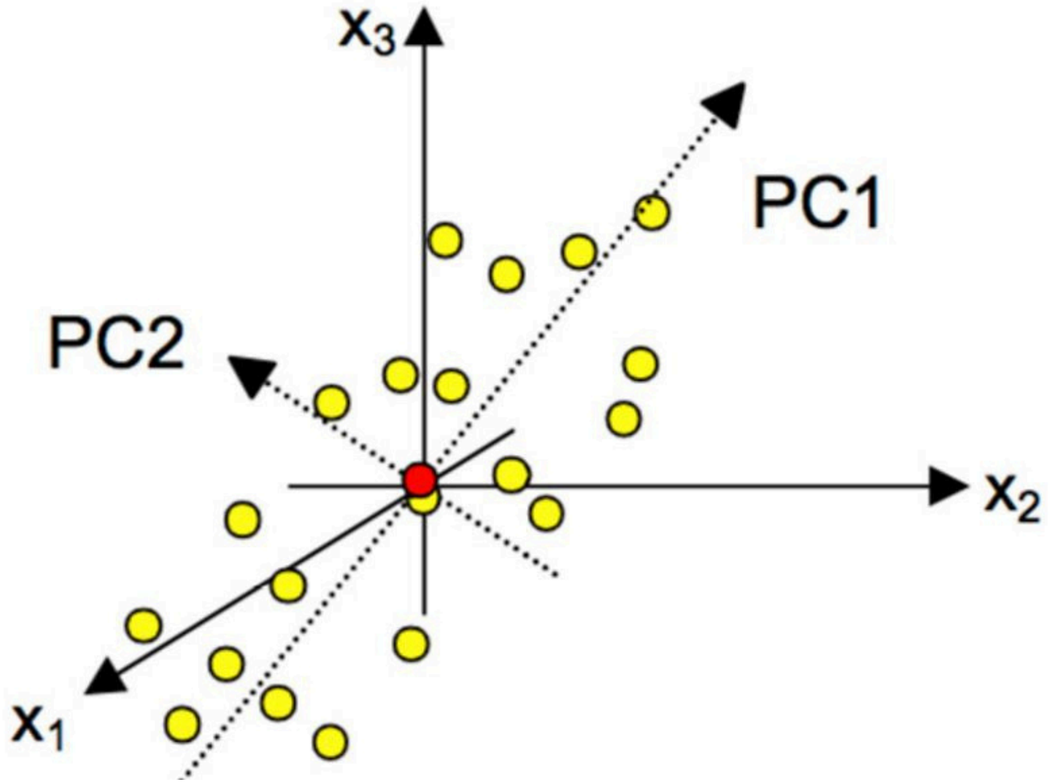


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Ordination/Dimensionality reduction

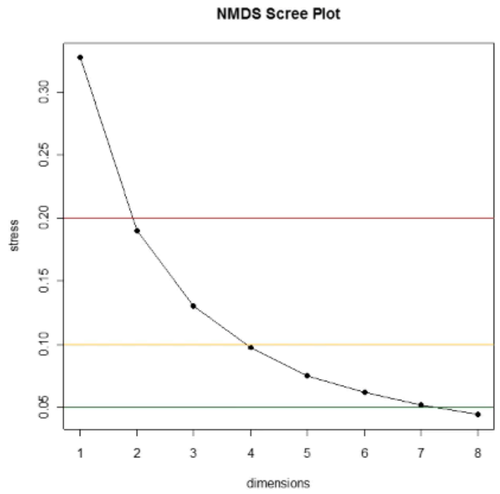
Non-metric Multidimensional Scaling (NMDS):

Multivariate data

	Sp A	Sp B	Sp C
Site 1	…	…	…
Site 2	…	…	…
Site 3	…	…	…

User-defined distance matrix (real space)

	Site 1	Site 2	Site 3
Site 1	…		
Site 2	…	…	
Site 3	…	…	…



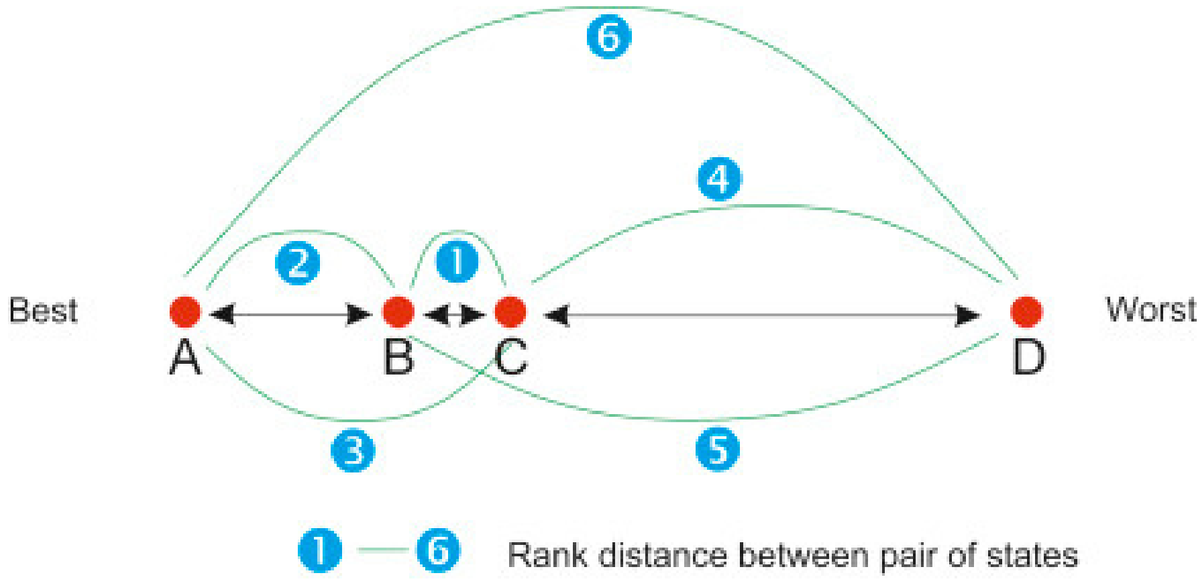
Goodness of fit:

>0.2 Poor (risks in interpretation)

0.1-0.2 Fair (some distances misleading)

0.05-0.1 Good (inferences confident)

<0.05 Excellent



PERMANOVA

Multivariate differential abundance testing: Statistical test applied to distance matrices (e.g., beta diversity distances).

Distance/Dissimilarity ~ Group (covariates)

Compartment	Variable	PERMANOVA		
		F. Model	R^2	*p*
L	Timepoint	2.978	0.090	0.006
	Root section	13.681	0.137	0.001
	Tree	3.637	0.292	0.001
	Timepoint * Root section	1.935	0.058	0.076
T	Timepoint	2.980	0.119	0.018
	Root section	7.259	0.096	0.001
	Tree	3.248	0.345	0.002
	Timepoint * Root section	1.680	0.067	0.112

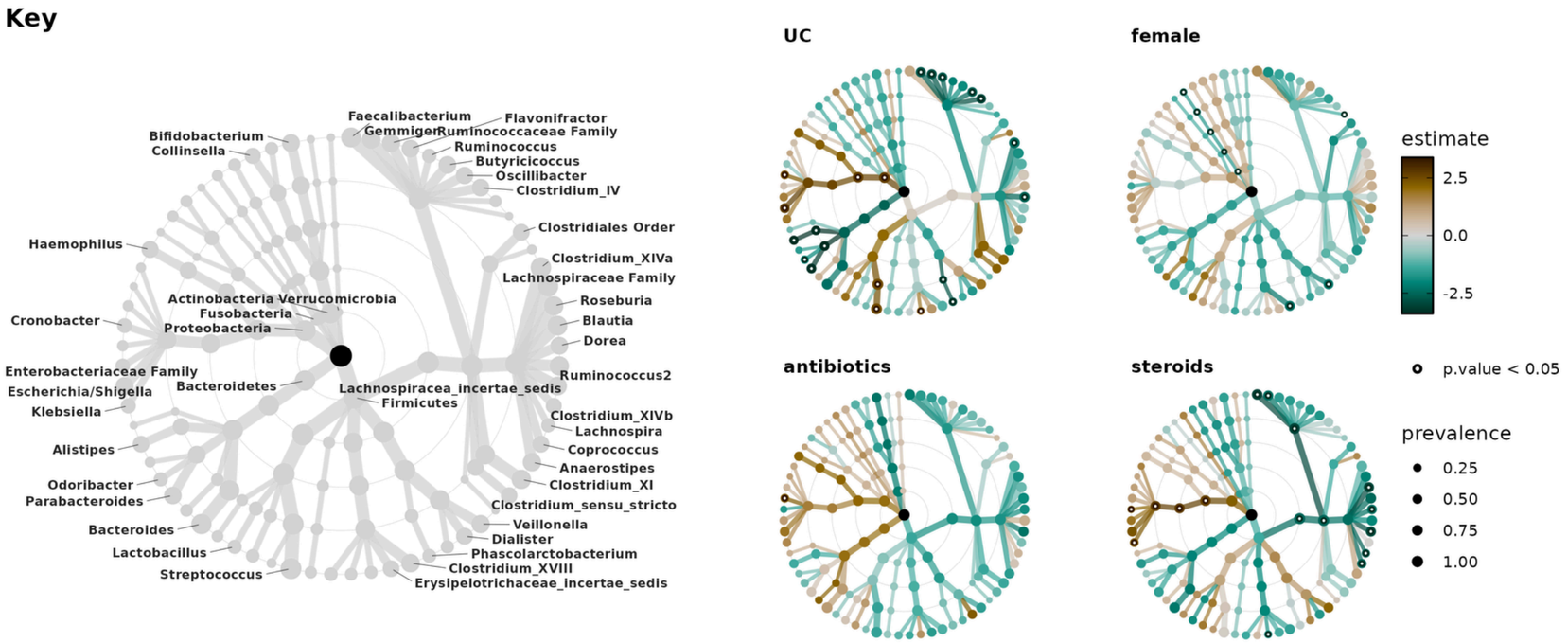
Becker et al. (2022)

Sensible to dispersion within-group dispersion

Statistical modeling

Statistical modeling: Applies models to identify relationships between microbial features and variables.

Taxa ~ Variables



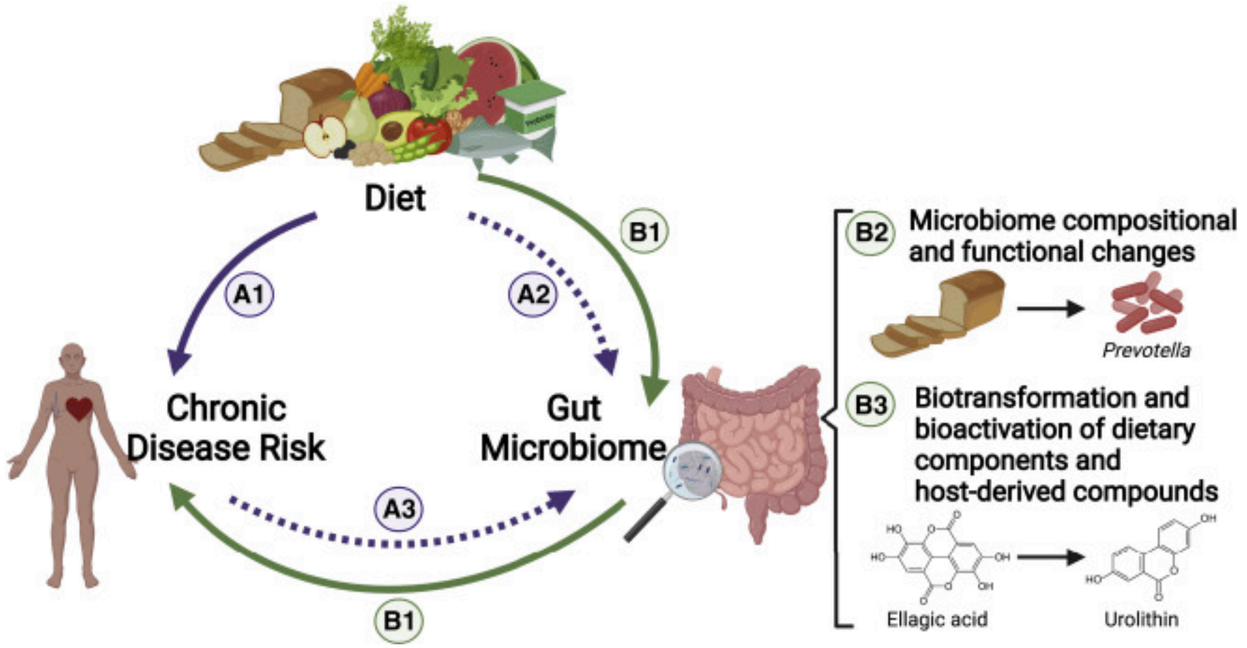
Linear Models or GLMs:

- ALDEx3
- ANCOM-BC2
- LinDA
- MaAsLin3

Statistical modeling

Statistical modeling: Predict phenotypic traits from microbial features (taxa).

Phenotype ~ Taxon abundance + covariates

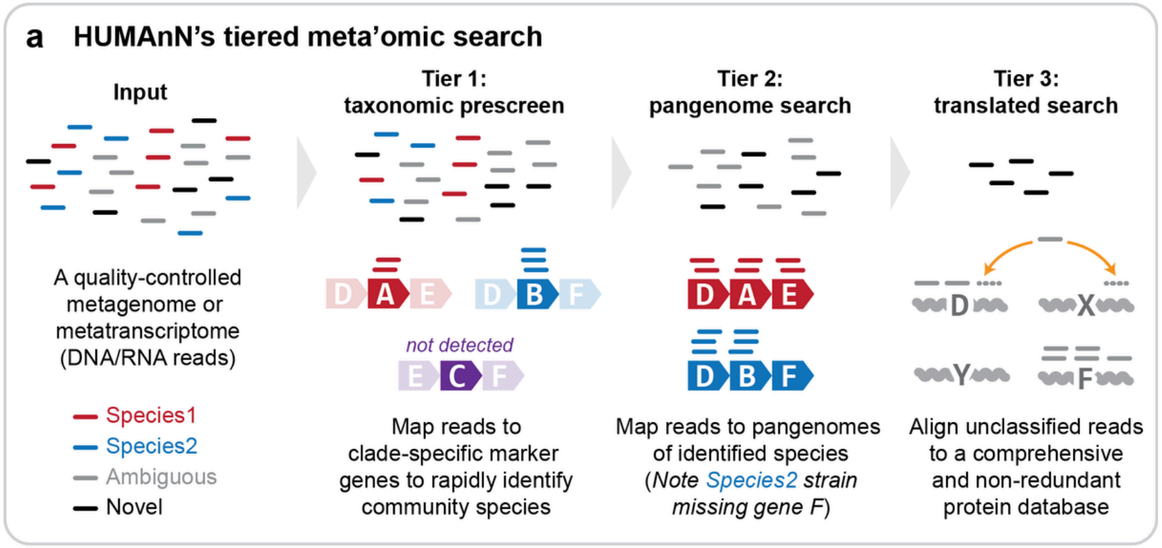


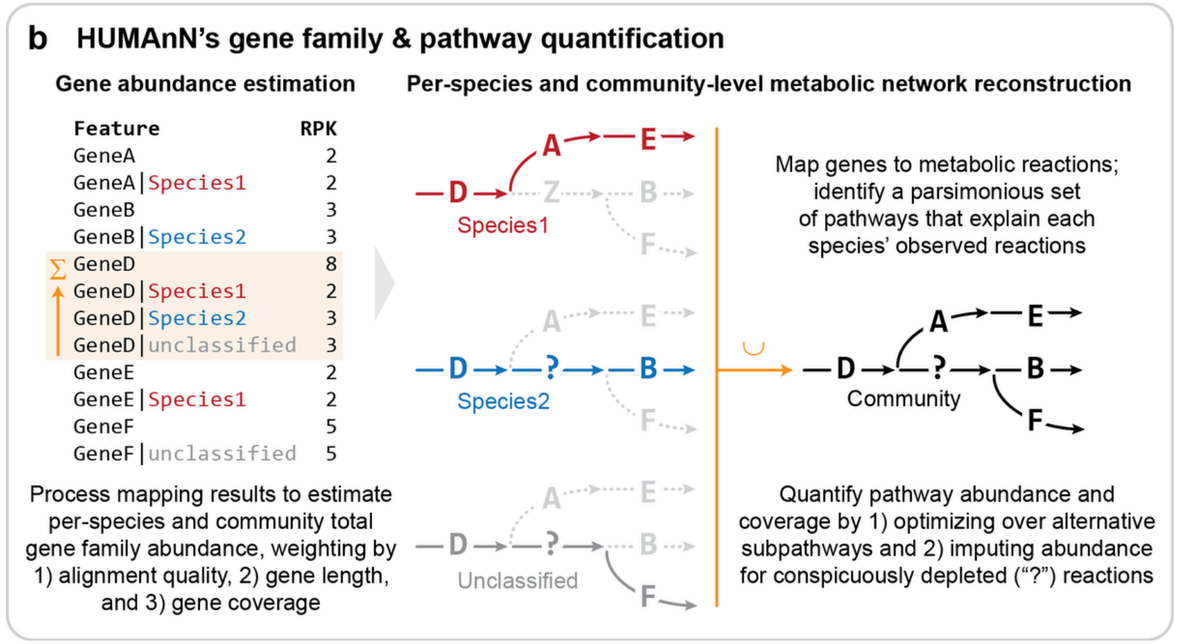
Machine learning:

- Random forest + CLR-transformed data
- SIAMCAT

Functional profiling

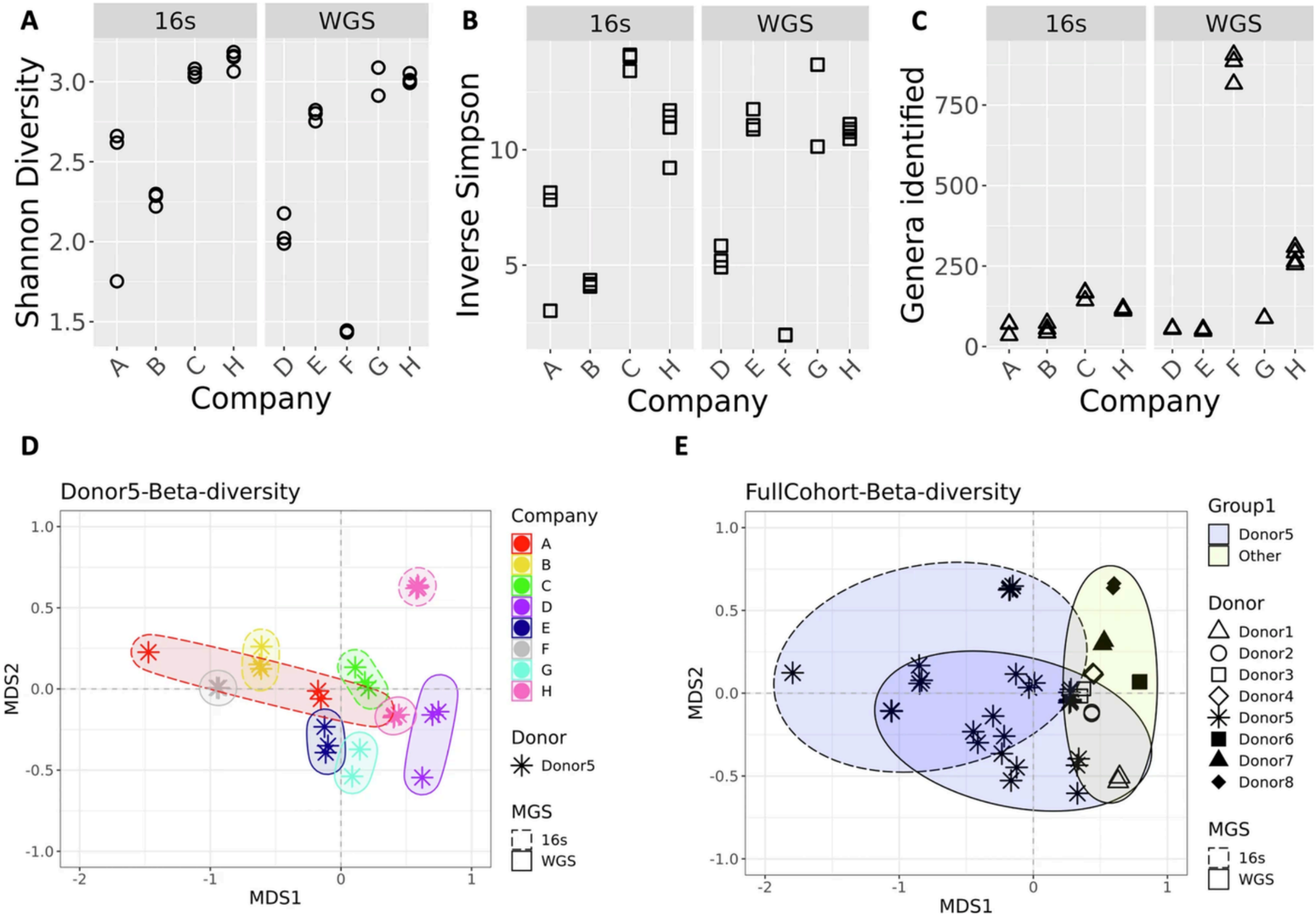
HUMAnN 3: Functionally profiles genes, pathways, and modules from metagenomes, now using native UniRef90 annotations from ChocoPhlAn species pangenomes.





Beghini et al. (2021)

Watch out with reproducibility!

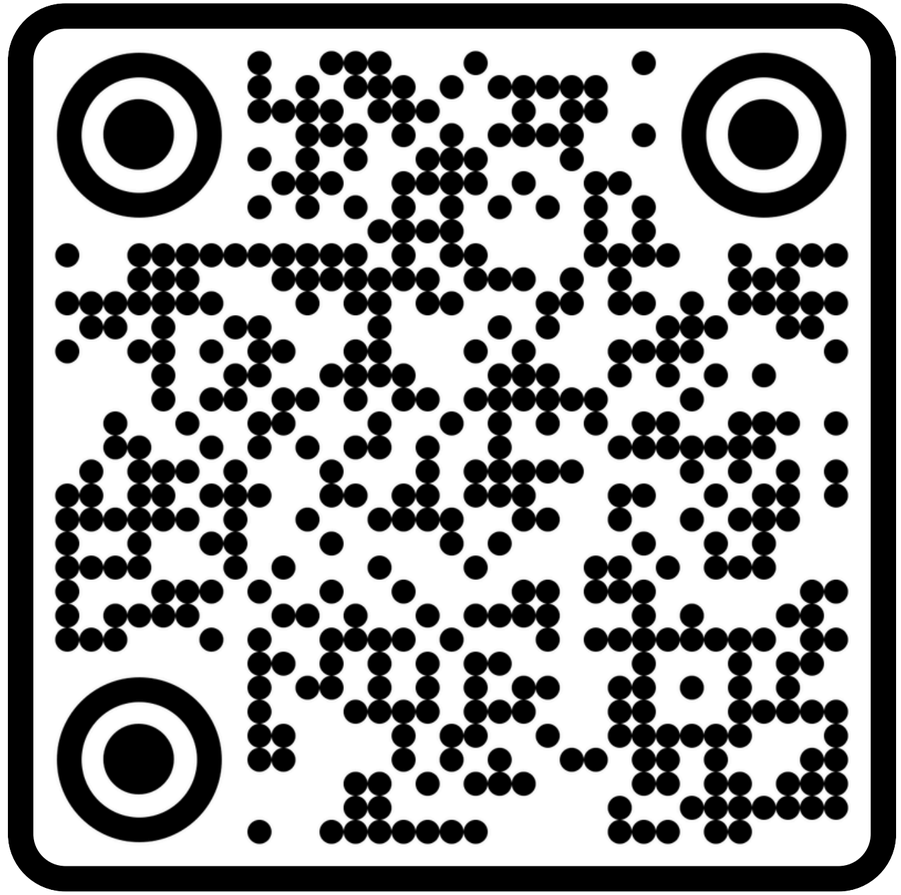


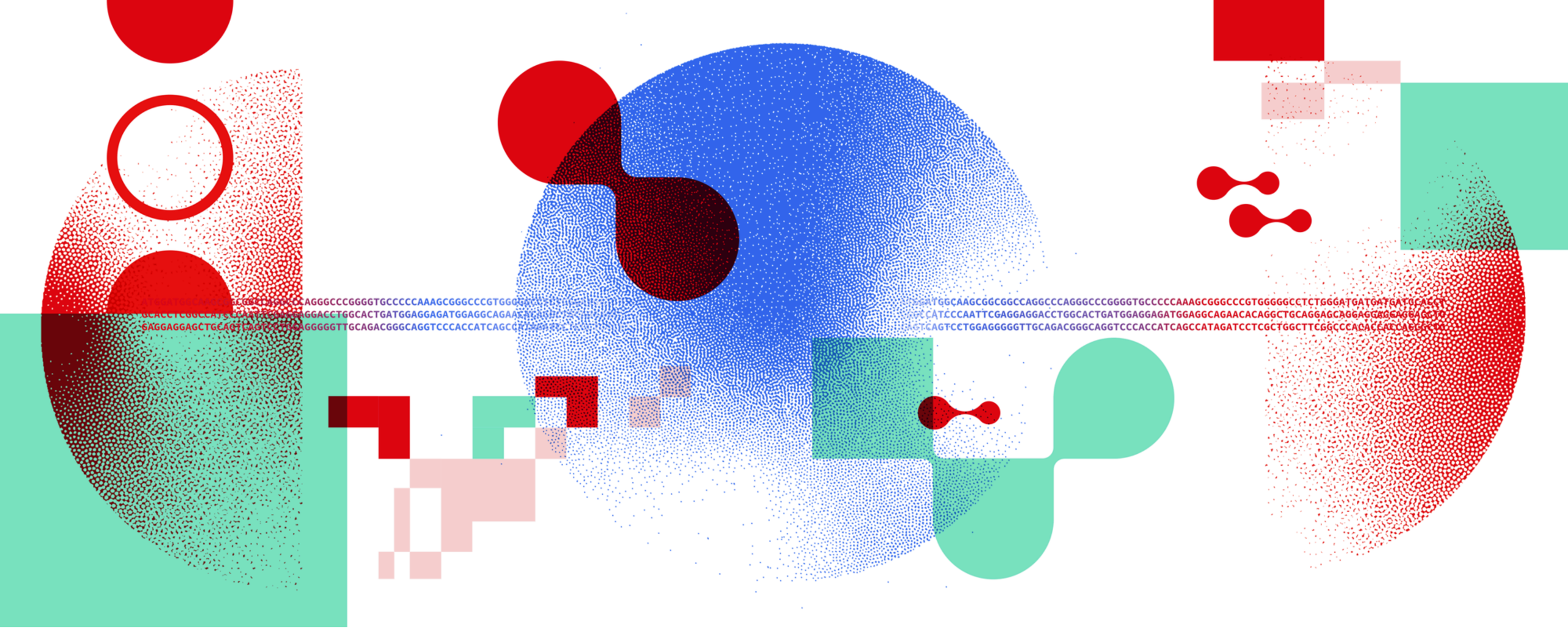
Servetas et al. (2026)

Summary

- There are several methods to characterize the community composition. It is advisable to verify the results with as many methodologies as possible.
- Use different metrics to corroborate initial or exploratory observations.
- “Microbiome datasets are compositional: And this is not optional ”.
- Be aware of the multiple existing methods to establish differentially abundant taxa, and the implications of using each of them.

Practicals





Thank you

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